

RUMANA FERDOUS MUNNE, Ph.D.

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PROFILE

NLP researcher specializing in LLM reasoning, document-level information extraction, entity alignment, and knowledge representation. My work studies how language models and structured knowledge can infer implicit meaning from complex, domain-specific documents and transform it into structured, interpretable, and useful representations. My current interests include scientific and biomedical NLP, ontology-guided reasoning, low-resource information extraction, and emotion-aware text generation for LLMs.

EDUCATION

- **Ph.D. in Informatics** — National Institute of Informatics, Tokyo, Japan (2023).
 - **M.Sc. in Informatics** — National Institute of Informatics, Tokyo, Japan (2020).
 - **M.Sc. in Computer Science & Engineering** — University of Dhaka, Bangladesh (2014).
 - **B.Sc. in Computer Science & Engineering** — University of Dhaka, Bangladesh (2011).
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EXPERIENCE

Postdoctoral Researcher — Artificial Intelligence Research Center (AIRC), AIST.
Language and Information Research Team (P.I.: Prof. Hiroya Takamura)

Apr 2026 – Present

- Study emotion understanding and emotion-aware text generation for LLMs, focusing on appraisal-style reasoning and response generation for emotionally sensitive applications.

Postdoctoral Researcher — RIKEN Center for Advanced Intelligence Project, Japan
Knowledge Acquisition Team (P.I.: Prof. Yuji Matsumoto)

Sep 2023 – Mar 2026

- Research and develop NLP methodologies for low-resource scientific and biomedical information extraction, focusing on implicit knowledge, ontology-guided verification, and domain-specific constraints.
- Contributed to multiple top-tier publications including EMNLP, COLING, SIGIR, TACL, EACL, ACL.

Research Assistant — National Institute of Informatics, Japan
Oct 2016 – Mar 2023

- Designed algorithms for knowledge graph alignment and completion, publishing findings at leading international conferences.

Special Researcher — Tokyo Institute of Technology, Japan
Apr 2022 – Jul 2023

- Led a collaborative research project between the National Institute of Informatics and the Tokyo Institute of Technology.

Senior Software Engineer — Samsung R&D Institute Bangladesh
Dec 2011 – Aug 2016

- Developed OS features for Samsung wearable devices and Galaxy series.
- Mentored team members and supported project commercialization.
- Participated in Samsung software showcasing worldwide.

Software Engineer — Samsung Inc., South Korea (OJT)
Mar 2013 – May 2013

- Collaborated on Android system applications for Galaxy smartphones.
 - Conducted engineering training and sessions.
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SELECTED PUBLICATIONS

- Munne, R.F., Rahman, M.M. and Matsumoto, Y., 2025 “Entity Profile Generation and Reasoning with LLMs for Entity Alignment.” **EMNLP 2025**.
 - Munne, R.F., Rahman, M.M. and Matsumoto, Y., 2025, July. “Lookalike Audience Expansion: A Graph-Based Model with LLMs.” **SIGIR eCom'2025**.
 - Munne, R.F., Nishida, N., Liu, S., Tokunaga, N., Yamagata, Y., Kozaki, K. and Matsumoto, Y., 2025, January. “Zero-Shot Entailment Learning for Ontology-Based Biomedical Annotation Without Explicit Mentions.” **COLING 2025**.
 - Liu, S., Nishida, N., Cheng, F., Tokunaga, N., Munne, R.F., Yamagata, Y., Kozaki, K., Utsuro, T. and Matsumoto, Y., “Better Generalizing to Unseen Concepts: An Evaluation Framework and an LLM-Based Auto-Labeled Pipeline for Biomedical Concept Recognition.” **EACL 2026**.
 - Nishida, N., Munne, R.F., Liu, S., Tokunaga, N., Yamagata, Y., Cheng, F., Kozaki, K., Matsumoto, Y. “Dissecting GraphRAG: A Modular Analysis of Knowledge Structuring for Factoid Question Answering.” Transactions of the Association for Computational Linguistics (**TACL**) **2026**.
 - El Khettari, O., Nishida, N., Liu, S., Munne, R.F., Yamagata, Y., Quiniou, S., Chaffron, S. and Matsumoto, Y. “Mention-Agnostic Information Extraction for Ontological Annotation of Biomedical Articles.” BioNLP, **ACL 2024**.
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SKILLS: Python, PySpark, Hive, TensorFlow, PyTorch, Neo4j, Graph Learning, NLP, LLM Reasoning, Information Extraction, Entity Alignment, Knowledge Graphs, C/C++, Java, SQL.

HONORS & AWARDS: Samsung R&D Iconic Team Member (2013), Young Leadership Award (2015), Monbukagakusho (MEXT) Scholarship (2016), University of Dhaka Outstanding Result. Scholarship (2011).